

Functional sequences, series and spaces – 3

Power series

13 сентября 2026 года

Theoretical minimum.

Theorem 1 (Cauchy–Hadamard). The series $\sum a_n x^n$ converges absolutely for $|x| < R$ and diverges for $|x| > R$, where $\frac{1}{R} = \limsup \sqrt[n]{|a_n|}$; the convergence is uniform on every $[-r, r]$ with $r < R$ and need not be uniform on $(-R, R)$. Differentiating or integrating term by term does not change R , so the sum is C^∞ inside the interval of convergence.

Theorem 2 (Abel). If $\sum a_n$ converges, then $f(x) = \sum a_n x^n$ tends to $\sum a_n$ as $x \rightarrow 1^-$. The converse is false — the Abel limit may exist for a divergent series — and this is precisely what makes Abel summation a summation method.

Theorem 3 (Tauber, Littlewood). If the Abel limit of $\sum a_n$ exists and $a_n = o(1/n)$, then $\sum a_n$ converges to that limit. Littlewood's theorem weakens the condition to $a_n = O(1/n)$; without any condition of this kind the implication fails. A condition that turns a summability statement back into convergence is called Tauberian.

Fact 1 (analytic versus smooth). A C^∞ function equals its Taylor series near a point if and only if the remainders tend to zero, which holds as soon as $|f^{(n)}| \leq C^n n!$ locally. Borel's theorem (Differentiable functions – 8) says that any sequence whatsoever occurs as the sequence of Taylor coefficients, so the class of analytic functions is very thin inside C^∞ .

Fact 2 (the boundary). A power series with radius R has at least one singular point on the circle $|x| = R$; if the coefficients are non-negative, the

point $x = R$ itself is singular (Pringsheim). A lacunary series such as $\sum x^{2^n}$ is singular at every point of the circle.

1. Prove the Cauchy–Hadamard formula, prove that term-by-term differentiation and integration do not change the radius, and compute the radius for $\sum \frac{n!}{n^n} x^n$, $\sum x^{n^2}$, $\sum \frac{x^n}{a^{\sqrt{n}}}$ and $\sum d(n)x^n$, where $d(n)$ is the number of divisors.
2. Prove Abel’s theorem, deduce $\sum \frac{(-1)^{n-1}}{n} = \ln 2$ and $\sum \frac{(-1)^n}{2n+1} = \frac{\pi}{4}$, and show by the example $\sum (-1)^n$ that the converse fails.
3. (Kapacëв 1.66) Prove that if $\sum_{n \geq 0} a_n$ is Cesàro summable, then $f(x) = \sum a_n x^n$ extends continuously from $[0, 1)$ to the point 1 (Frobenius’ theorem), and (Kapacëв 1.62) that Cesàro summability forces $\frac{a_1 + \dots + a_n}{n} \rightarrow 0$.
4. Prove Tauber’s theorem, state Littlewood’s strengthening, and show that the Abel limit of a divergent series can be any prescribed number, so that no conclusion at all is possible without a Tauberian hypothesis.
5. (Kapacëв 1.65) Prove that a function infinitely differentiable on an interval, all of whose derivatives are non-negative, is real-analytic there.